HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Assisted in the evaluation and selection of **materials** for new product prototypes, focusing on performance characteristics and cost-effectiveness.
- Conducted **material testing and analysis** using established laboratory procedures to assess durability and ensure compliance with product specifications, documenting results in detailed reports.
- Supported investigations into material defects and quality concerns, analysing test data to identify root causes and formulating corrective action recommendations for the manufacturing team.
- Collaborated with **manufacturing and quality teams** to integrate new material specifications into production workflows, aiming to enhance product performance and streamline efficiency by approximately 15%.
- Prepared comprehensive **test result analyses** and presented findings, including actionable recommendations, to stakeholders across product development and senior management.
- Assisted in the **evaluation of suppliers** and the qualification of new material components, ensuring they met required industry standards and internal quality benchmarks.
- Maintained detailed **compliance documentation** for materials used, ensuring adherence to relevant industry standards and internal quality management system requirements.

SKILLS

Materials Science – Material Selection, Material Testing, Material Analysis, Failure Analysis, Quality Assurance, Supplier Evaluation

Engineering – Product Development, Technical Input, Process Improvement, Continuous Improvement, Root Cause Analysis, Compliance

Data Analysis & Reporting – Test Result Analysis, Data Analysis, Reporting, Presentations, Stakeholder Communication, Data Interpretation

Software & Tools – Jira, Confluence, Microsoft Office Suite (Word, Excel, PowerPoint)

Laboratory Skills – Material Characterization, Durability Testing, Performance Assessment, Specification Compliance

PROJECTS

Composite Material Performance Analysis (HCL Technologies)

- Performed comparative analysis on various **composite materials** for an enterprise client's product line, evaluating tensile strength and impact resistance under simulated environmental conditions.
- Documented findings in a structured report, highlighting material suitability for specific application requirements and providing data-driven recommendations for selection.
- Collaborated with the client's engineering team to interpret test results and align material choices with project goals, ensuring adherence to performance metrics.

Product Quality Improvement Initiative (HCL Technologies)

- Investigated recurring quality concerns with a specific component in an enterprise operations dashboard, analysing production logs and failure reports.
- Developed and executed a series of targeted **material tests** to pinpoint degradation factors, identifying a critical weakness in the chosen polymer under stress.
- Recommended an alternative material with enhanced durability, leading to a projected reduction in component failure rates by an estimated 20% for the operational system.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)